

WEEKLY REPORT :

Project Title: Self Driving RC Car

Date: 10/6/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

Project Members:

Ethan Wells (Group Leader)

Alexis Perez

Ethan Bishop

Abigail Duran

Weekly Summary (10/6/2026):

- This week the team moved from preparing US traffic-sign data to testing a model and narrowing hardware choices for the vehicle. Ethan Wells completed last week's goal of training and testing a CNN on the prepared 12-class LISA dataset. It reached 89.90% accuracy on 584 held-out test crops. He reviewed the classes with lower accuracy and prepared 3,449 Mapillary sign crops for the next training run. Alexis Perez selected and ordered line sensors to help the car follow the course, and completed an OpenCV beginner course. Ethan Bishop found 3D-printable chassis plate files and continued researching whether ROS 2 would fit the vehicle's software. Abigail Duran organized the components and connections needed for the custom PCB and reviewed the Compute Module 5 IO Board to determine which features to keep. PCB layout work carries into next week.

Proposed Plan for Next Week:

- Ethan Wells: Train and test a CNN on the prepared Mapillary sign data; if time allows, combine the LISA and Mapillary datasets and test a model trained on both.
- Alexis Perez: Connect the line sensors to the Raspberry Pi and begin using OpenCV with the Raspberry Pi camera.

- Ethan Bishop: Continue researching ROS 2 and assess the proposed camera, sign detector, behavior, and motor driver nodes for the vehicle.
- Abigail Duran: Begin the PCB layout, including component placement, board dimensions, power distribution, and signal routing; transfer the selected components into the design.

Weekly Contributions:

- Ethan Wells:
 - 10/2/2026 – Trained the CNN on the prepared 12-class LISA sign dataset (2 hours)
 - 10/5/2026 – Tested the LISA model on the held-out test set and reviewed the per-class results (2 hours)
 - 10/6/2026 – Prepared 3,449 Mapillary sign crops for the same 12 classes (3 hours)
- Alexis Perez:
 - Selected and ordered line sensors for course following
 - Completed an OpenCV beginner course
- Ethan Bishop:
 - Researched 3D-printable chassis plate files
 - Continued researching a possible ROS 2 implementation for the vehicle
- Abigail Duran:
 - Organized the components and connections needed for the custom PCB
 - Reviewed the Compute Module 5 IO Board to determine which features to keep in the custom design

Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	7	33
Alexis Perez	6	29
Ethan Bishop	6	28
Abigail Duran	6	29